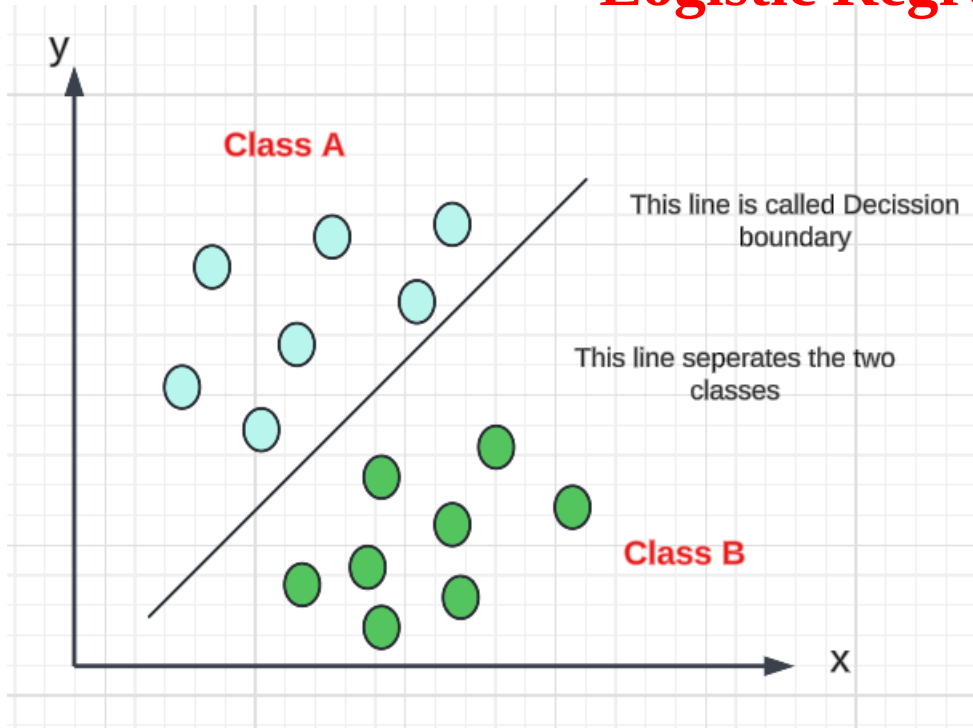


One vs Rest

- **Logistic Regression One-vs-Rest (OvR)**, also called **One-vs-All (OvA)**
- **Logistic regression** is inherently a **binary classifier**, meaning it **predicts two outcomes**. However, **many real-world problems** involve **more than two classes**.
- **One-vs-Rest (OvR)** is a method used to **adapt binary logistic regression** for these **multiclass problems**.

Logistic Regression-Binary Classification

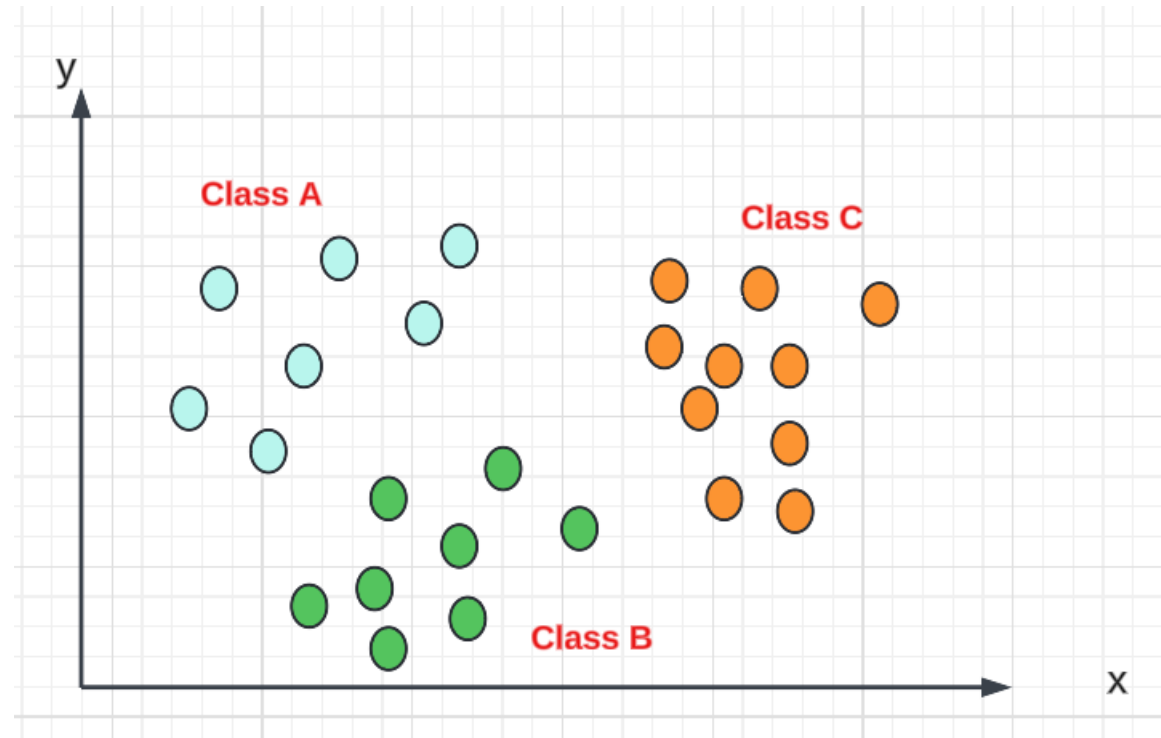


In **binary classification**, **logistic regression** creates a **decision boundary** instead of a simple line.

If the **logistic regression outputs** a **probability of 0.5**, this is often used as the threshold to **classify data** into **one of two categories (e.g., 0 or 1)**.

In a 2D feature space, this **decision boundary** could appear as a **straight line** or a **hyperplane** in higher dimensions that **separates the two classes**.

Multi class Classification



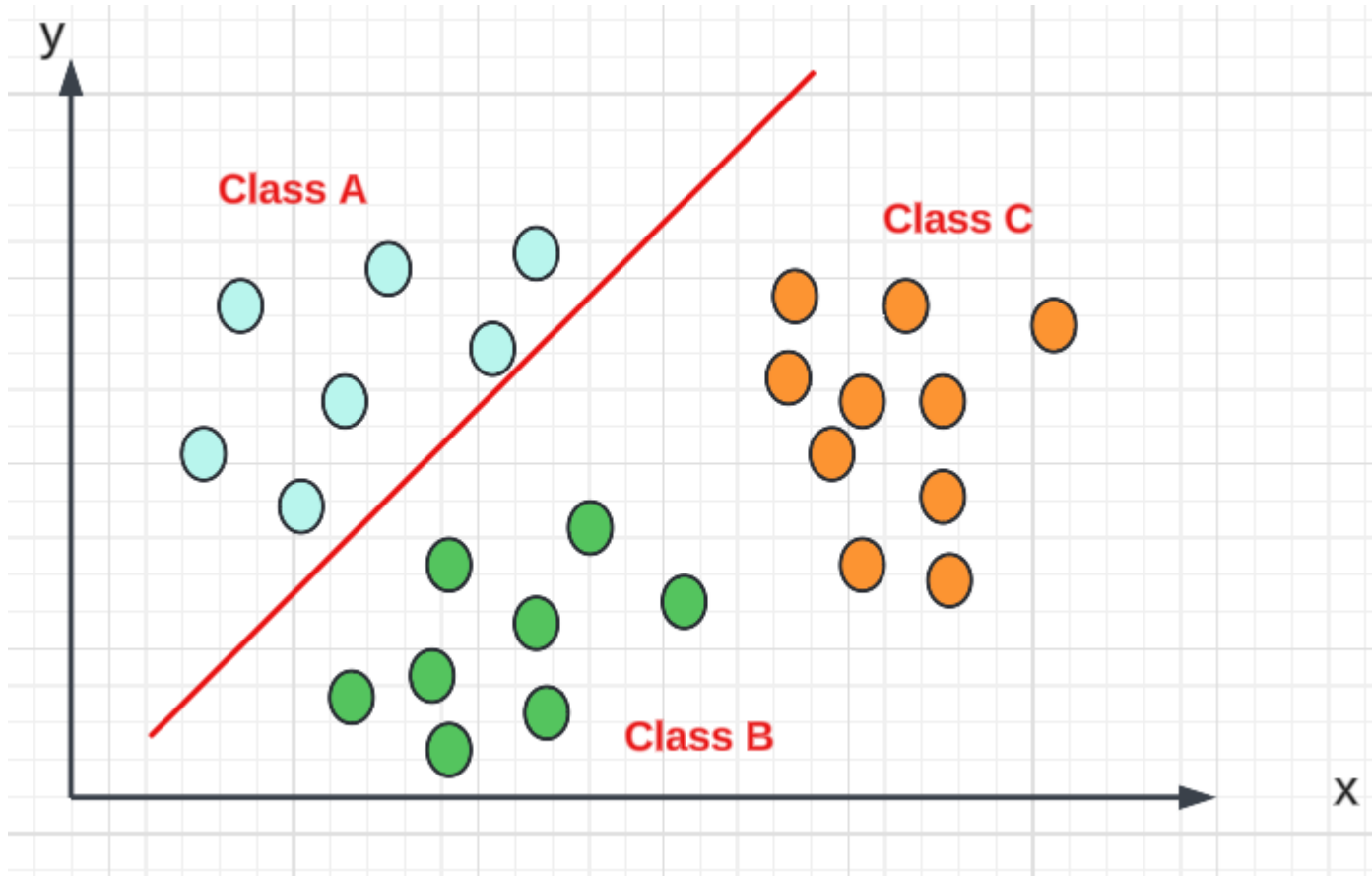
To solve this **Multi Class Classification Problem** we can use **One vs Rest** version of **Logistic Regression**

X1	X2	X3	O/P
A	B	F	O1
B	C	F	O2
C	B	F	O1
H	G	J	O3
K	K	I	O2
k	R	U	O3
U	H	Y	O2

In this **Dataset** We have **3** different
Classes of Output
O1,O2,O3

One vs Rest – Logistic Regression

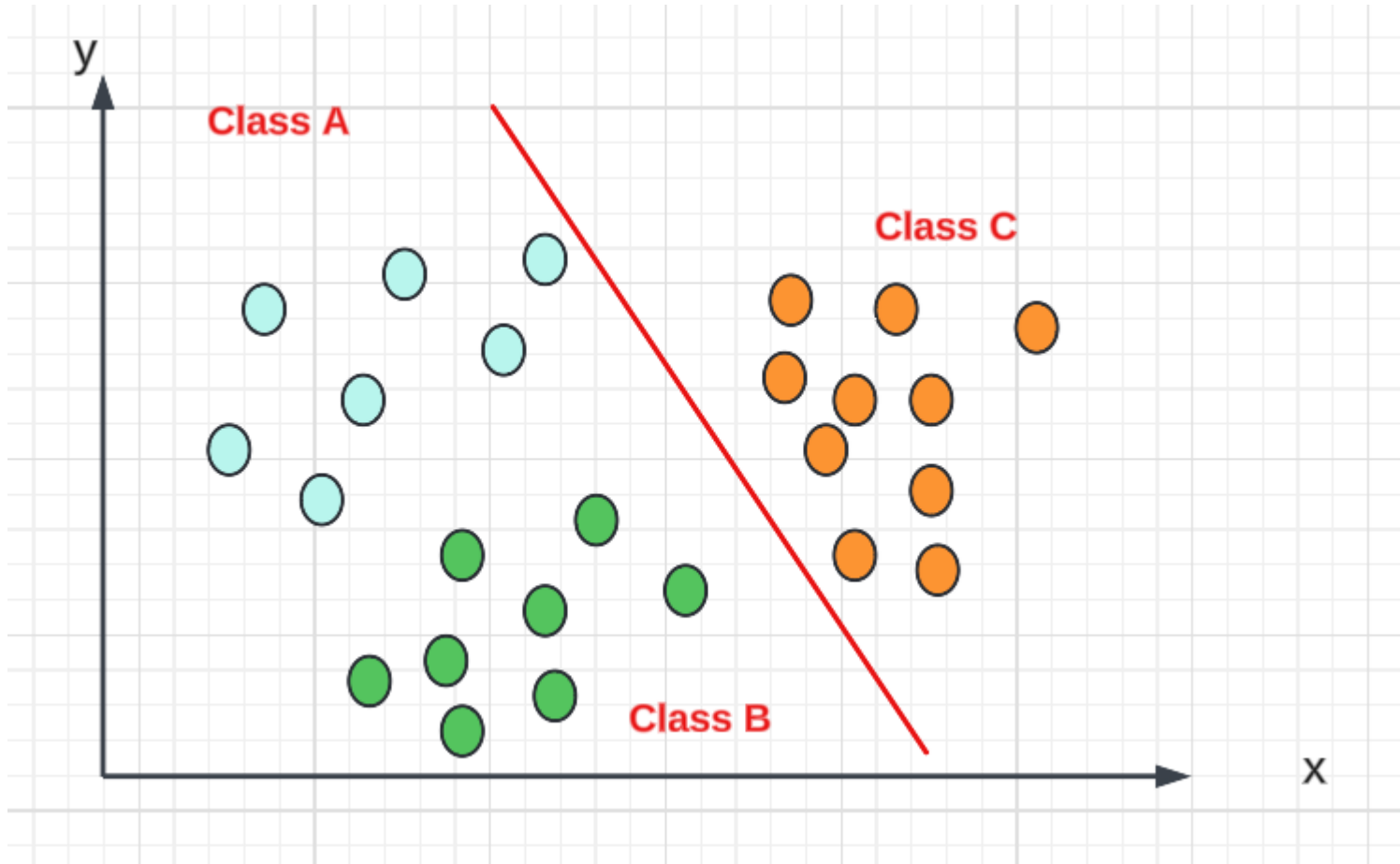
- In the **OvR** approach, **a separate logistic regression model** is trained for **each class**.
- For a problem with **N classes**, **N binary classifiers** are created.



Here – Binary Classification

Model M1 --- Classifies **Class A** as **Positive Class**

Remaining **Class B** and **Class C** as **Negative class**



Here Binary Classification

Model M2 --- Classifies **Class C** as **Positive Class**

Remaining **Class A** and **Class B** as **Negative class**



Here Binary Classification

Model M3 --- Classifies **Class B** as **Positive Class**

Remaining **Class A** and **Class C** as **Negative class**

One –Hot Encoding in the dataset

X1	X2	X3	O/P	O1	O2	O3
A	B	F	O1	1	0	0
B	C	F	O2	0	1	0
C	B	F	O1	1	0	0
H	G	J	O3	0	0	1
K	K	I	O2	0	1	0
k	R	U	O3	0	0	1
U	H	Y	O2	0	1	0

X1	X2	X3	O/P	O1	O2	O3
A	B	F	O1	1	0	0
B	C	F	O2	0	1	0
C	B	F	O1	1	0	0
H	G	J	O3	0	0	1
K	K	I	O2	0	1	0
k	R	U	O3	0	0	1
U	H	Y	O2	0	1	0

Model

M1 will take **x1 ,x2,x3** as input
 Features **O1** as the output Feature

X1	X2	X3	O/P	O1	O2	O3
A	B	F	O1	1	0	0
B	C	F	O2	0	1	0
C	B	F	O1	1	0	0
H	G	J	O3	0	0	1
K	K	I	O2	0	1	0
k	R	U	O3	0	0	1
U	H	Y	O2	0	1	0

Model

M2 will take **x1 ,x2,x3** as input

Features O2 as the **output Feature**

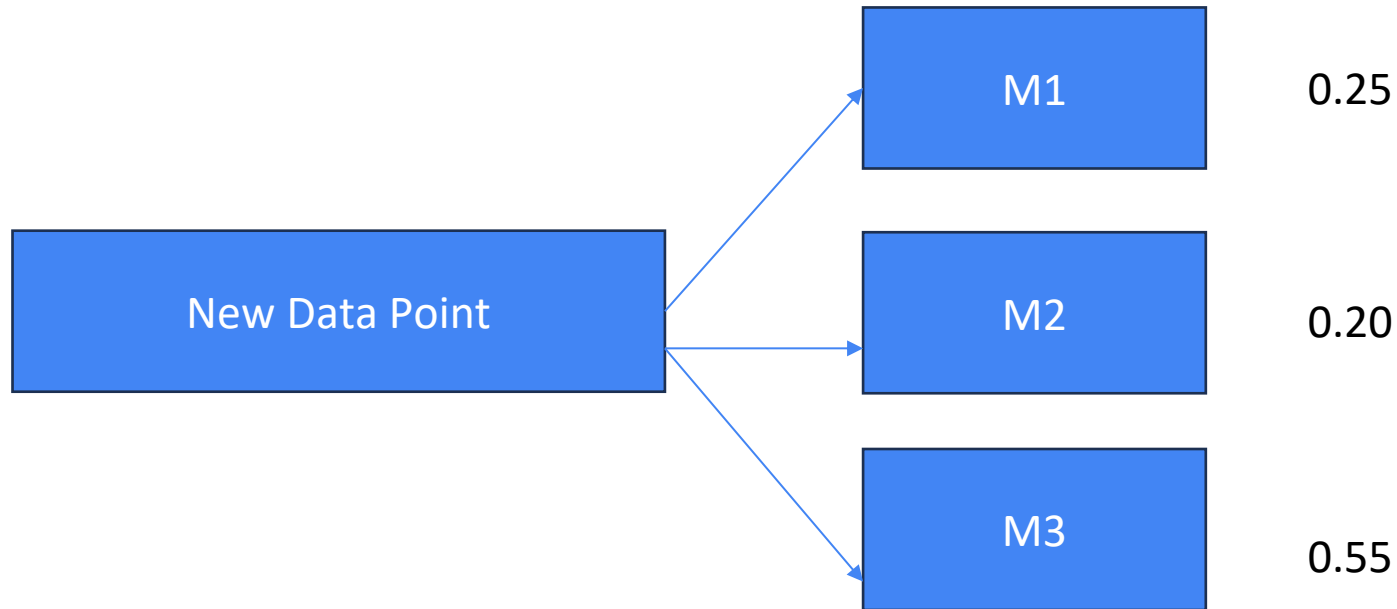
X1	X2	X3	O/P	O1	O2	O3
A	B	F	O1	1	0	0
B	C	F	O2	0	1	0
C	B	F	O1	1	0	0
H	G	J	O3	0	0	1
K	K	I	O2	0	1	0
k	R	U	O3	0	0	1
U	H	Y	O2	0	1	0

Model

M3 will take **x1 ,x2,x3** as input

Features O3 as the **output Feature**

Prediction of New data point



For a New Data point

- Each of the **Model** will gives a **Score or probability** that the **data point belongs to its class**
- If the **Model-M3** for **O3 Class** gives the **highest probability**, then the data point belongs to **O3 Class**.